

Maxima and Minima

1. Consider the following functions:

i. $y = \frac{1}{3}x^3 + \frac{1}{2}x^2 - 6x + 8$

ii. $y = x^4 + 2x^3 - 3x^2 - 4x + 4$

iii. $y = 3x^4 - 10x^3 - 12x^2 + 12x - 7$

iv. $y = 3x^4 - 10x^3 - 12x^2 + 12x - 7$

v. $y = x^4 - 6x^3 + 12x^2 - 8x$.

Find the following:

- (i) the critical points
- (ii) the intervals on which y is increasing and decreasing
- (iii) the maximum and minimum values of y
- (iv) the points of inflection
- (v) the equations of the tangents at the points of inflection

Finally, sketch the graph of $f(x)$.

2. Find the critical numbers of f if $f(x) = (x+5)^2 \sqrt[3]{x-4}$.

3. Given $f(x) = x^3 + x^2 - 5x - 5$.

- (i) Find the intervals on which f is increasing and the intervals on which f is decreasing.
- (ii) Find the intervals on which the graph of f is concave upward and concave downward.
- (iii) Sketch the graph of f .

4. If $f(x) = 12 + 2x^2 - x^4$, find the maximum and minimum values of f . Discuss concavity, find the points of inflection and sketch the graph of f .

5. Find the equations of the tangents at the points of inflection of

$$f(x) = x^4 - 6x^3 + 12x^2 - 8x$$

Sketch the graph of the function.

6. Find the maximum volume of a right circular cylinder that can be inscribed in a cone of altitude 12 cm and base radius 4 cm, if the axes of the cylinder and cone coincide.

7. A cylindrical container with circular base is to hold 64 cubic inches. Find the dimensions so that the amount of metal required is a minimum.

8. A circular cylindrical container, open at the top and having a capacity of 24π cubic inches, is to be manufactured. If the cost of the material used for the bottom of the container is three times that used for the curved part and if there is no waste of material, find the dimensions which will minimize the cost.

9. A rectangular box with length 3 times of its breadth is to have a volume of 288 cm^3 . Find its least possible surface area.

10. A sheet of paper for a poster contains 18 square feet. The margins at the top and bottom are 9 inches and at the sides 6 inches. What are the dimensions if the printed area is a maximum?

11. Find the dimensions of a cylindrical closed can with volume 64 cubic inches so that the amount of metal required is minimum.

12. Find the radius and height of the right circular cylinder of largest volume that can be inscribed in a right circular cone with radius 6 inches and height 10 inches.
13. Find a point on the curve $y = x^2$ that is closest to the point (18, 0).
14. Find the dimensions of the rectangle of maximum area that can be inscribed in the portion of the parabola $y^2 = 4px$ intercepted by the line $x = a$.
15. A garden is to be laid out in a rectangular area and protected by a chicken wire fence. What is the largest possible area of the garden if only 100 running feet of chicken wire is available for the fence?
16. A closed cylindrical can is to hold 1 liter (1000 cc) of liquid. How should we choose the height and radius to minimize the amount of material needed to manufacture the can?
17. Find the dimensions of the rectangle with maximum area that can be inscribed in a circle of radius 10.
18. Find the dimensions of the right circular cylinder of largest volume that can be inscribed in a sphere of radius R.